System Testing Report

Project 6: Factors Influencing Student Performance with SQL

Project Information

Responsible: Member 6 (UI/UX + Deployment)
Date: April 29, 2026
Status: Final Review

1. Testing Strategy

The testing strategy focuses on ensuring data integrity, API reliability, and analytical accuracy. We employ a mix of automated and manual testing to validate the student performance analysis system.

Testing Levels:

- **Unit Testing:** Validating individual SQL queries and API middleware components.
- **Integration Testing:** Ensuring the Node.js backend correctly interfaces with the PostgreSQL database.
- **Performance Testing:** Stress testing the analytics engine with datasets up to 395 records.
- **User Acceptance Testing (UAT):** Verifying that the dashboard meets the needs of educational stakeholders.

2. Test Execution Log

Test ID	Description	Requirement	Expected Result	Status
TS-01	CSV Import (395 records)	FR-01	Data imported < 30s	PASS
TS-02	Duplicate Data Entry	FR-02	Duplicates rejected	PASS
TS-03	Analytical Query Execution	FR-03	Results in < 2s	PASS
TS-04	RBAC: Unauthorized Access	FR-06	401 Unauthorized	PASS
TS-05	Filter Logic (Absences)	FR-04	Accurate student list	PASS

3. API Testing Summary

Using automated tools, the following endpoints were validated for response time and payload accuracy:

- POST /api/auth/login: Successfully issues JWT tokens for valid credentials.
- GET /api/analytics/correlation: Verified SQL join logic between student, family, and grade tables.
- Error Handling: Confirmed 404 for missing resources and 400 for malformed JSON payloads.

4. Final Recommendations

The system is stable and meets all functional requirements. Deployment to the AWS/Docker environment is cleared for the final submission.